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Data Science for Business and Economic Decisions
2/21/2025

Project Proposal

I plan to apply two techniques from chapter 2: linear regression and logistic regression. I would like to look at weather data, particularly temperature, rainfall and extreme weather events. Two datasets that I plan to use are NOAA's Daily Summary and NOAA's Storm Events Database. Each city's Daily Summary includes data on rainfall, maximum temperature, and minimum temperature. I would download the Daily Summary for a number of cities and combine them into one large dataset. The Storm Events Database reports the dates and states in which Tornadoes occurred.

With this data, I would like to run regressions on maximum temperature and minimum temperature with city fixed effects and a time trend in order to determine whether the average maximum and average minimum temperatures are changing over time. It would also be interesting to generate the monthly variance (across all days within a given month) of each temperature measure in a given city and run this same regression with the variance of each temperature measure instead of the temperature in order to see if temperatures are becoming more variable. I would do a similar thing for rainfall. I would also do a logit of extreme weather (such as tornadoes) with state fixed effects and a time trend in order to determine whether extreme weather is becoming more common. I would use these results to determine whether patterns in these variables offer evidence of climate change in the US.